

Fiber Reinforced Concrete (FRC)



- FRC is defined as concrete made with hydraulic cement, containing fine or/and coarse aggregate, and discontinuous discrete fibers
- Discrete fibers added to improve cracking resistance, toughness, fatigue life, impact resistance, or other engineering properties
- The fibers can be made from natural material or are a manufactured product such as glass, steel, carbon, and polymer

Typical Properties of Fibers

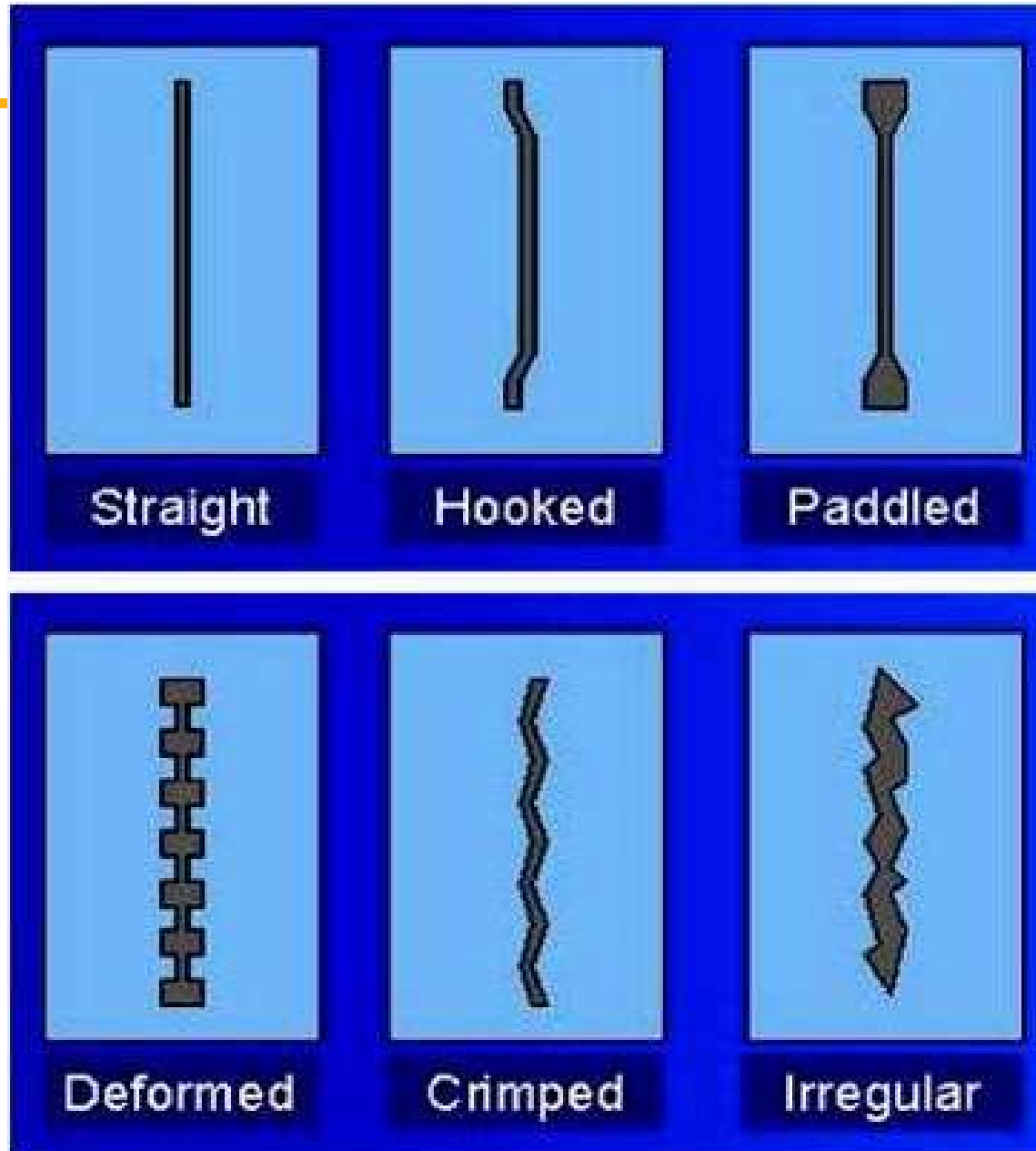


Type of fiber	Tensile strength (ksi)	Young's modulus (10^3 ksi)	Ultimate elongation (%)	Specific gravity
Acrylic	30-60	0.3	25-45	1.1
Asbestos	80-140	12-20	~0.6	3.2
Cotton	60-100	0.7	3-10	1.5
Glass	150-550	10	1.5-3.5	2.5
Nylon (high tenacity)	110-120	0.6	16-20	1.1
Polyester (high tenacity)	105-125	1.2	11-13	1.4
Polyethylene	~100	0.02-0.06	~10	0.95
Polypropylene	80-110	0.5	~25	0.90
Rayon (high tenacity)	60-90	1.0	10-25	1.5
Rock wool (Scandinavian)	70-110	10-17	~0.6	2.7
Steel	40-400	29	0.5-35	7.8

Source: ACI Committee 544, Report 544.1R-82, Concr. Int., Vol. 4, No. 5, p. 11, 1982

$$1 \text{ ksi} = 6.89 \text{ MPa}$$

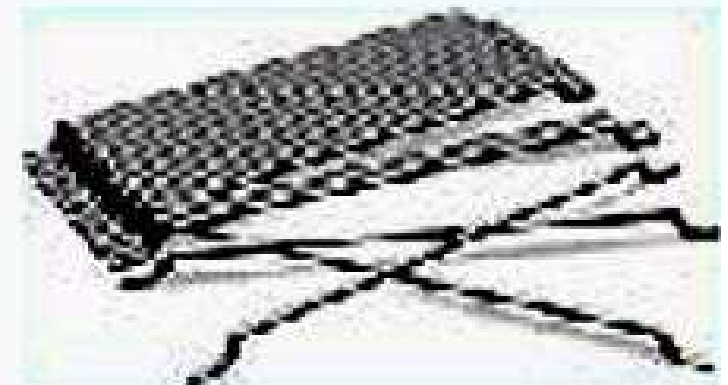
Types of Steel Fibers



Source: Kamran M. Nemati



Steel fibers glued together prior to mixing



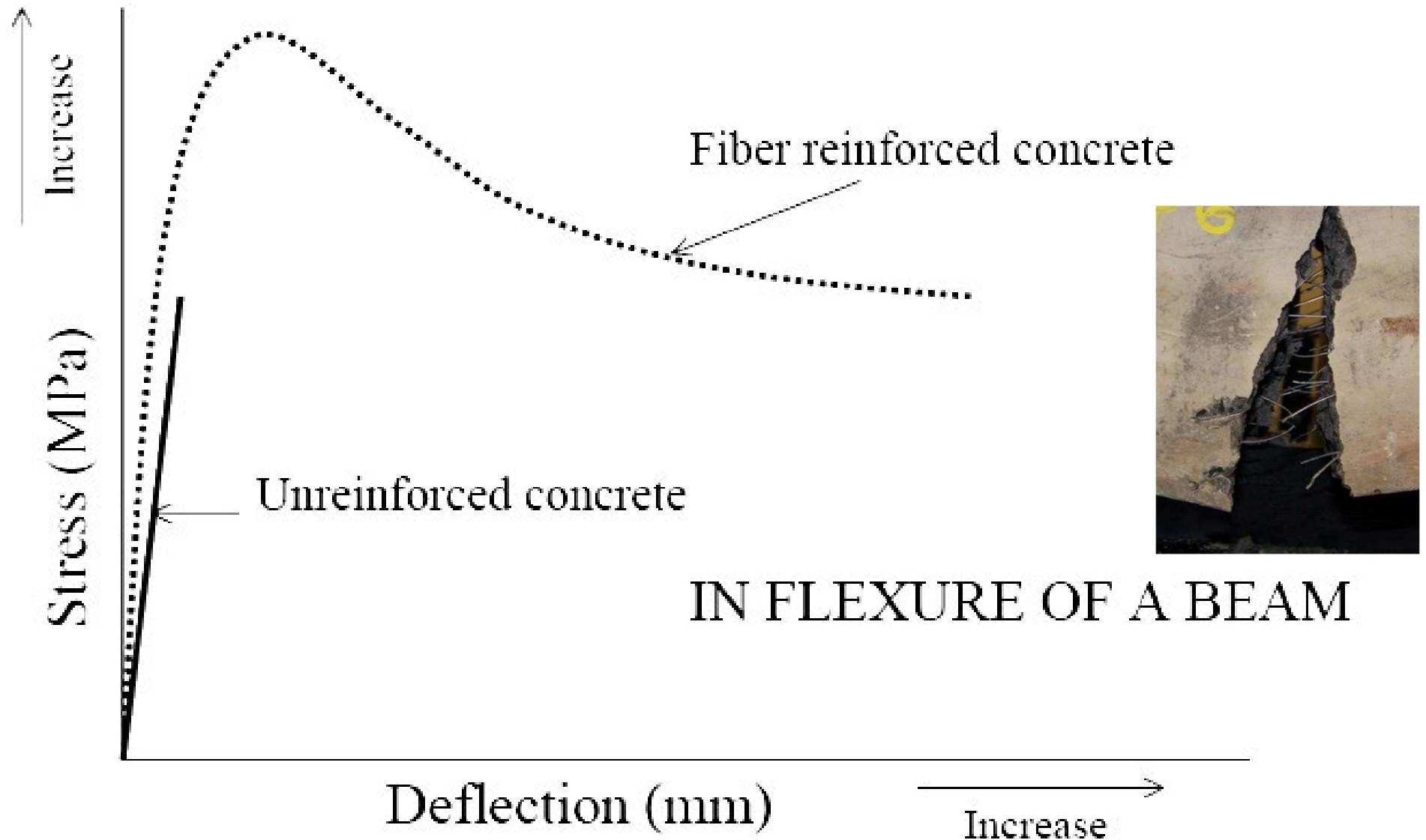
Separation of fibers occurs during mixing to ensure uniform distribution



Fiber Reinforced Concrete

- Generally, the fibers are **not added to increase the concrete strength**, though modest increases in strength may occur
- The principal role of the fibers is **to control the post-cracking behavior of concrete in tension**. Once the concrete matrix has cracked, fibers provides bridging across the cracks as they begin to open and transmit stress across a cracked section.
- Fibers provide *post-cracking ductility (i.e. toughness)* to the concrete.

Flexural Behavior of FRC vs Plain Concrete





Fiber Reinforced Concrete

- The workability of fiber reinforced mixes decreases as the fiber content increases and as the aspect ratio (l/d) increases
- Fiber volumes currently used in practice
 - Steel: $< 1.0\%$
 - Polypropylene: $< 0.5\%$
- High fiber volume ($> 3\%$) may be achieved through special processing techniques (high cost and impractical)
- FRC has been used for floor slabs, pavements, slope stabilization, and tunnel linings

Mix Proportions (Plain Concrete vs FRC in lb/yr³)



Material	Plain Concrete	FRC
Cement	752	875
Water (W/C = 0.45)	338	394
Fine aggregate	1440	1282
Coarse aggregate	1150	1024
Fibers (2% by volume)	-	265

Source: A.N. Hanna, PCA Report RD 049.01P, Portland Cement Association, Skokie, Ill., 1977

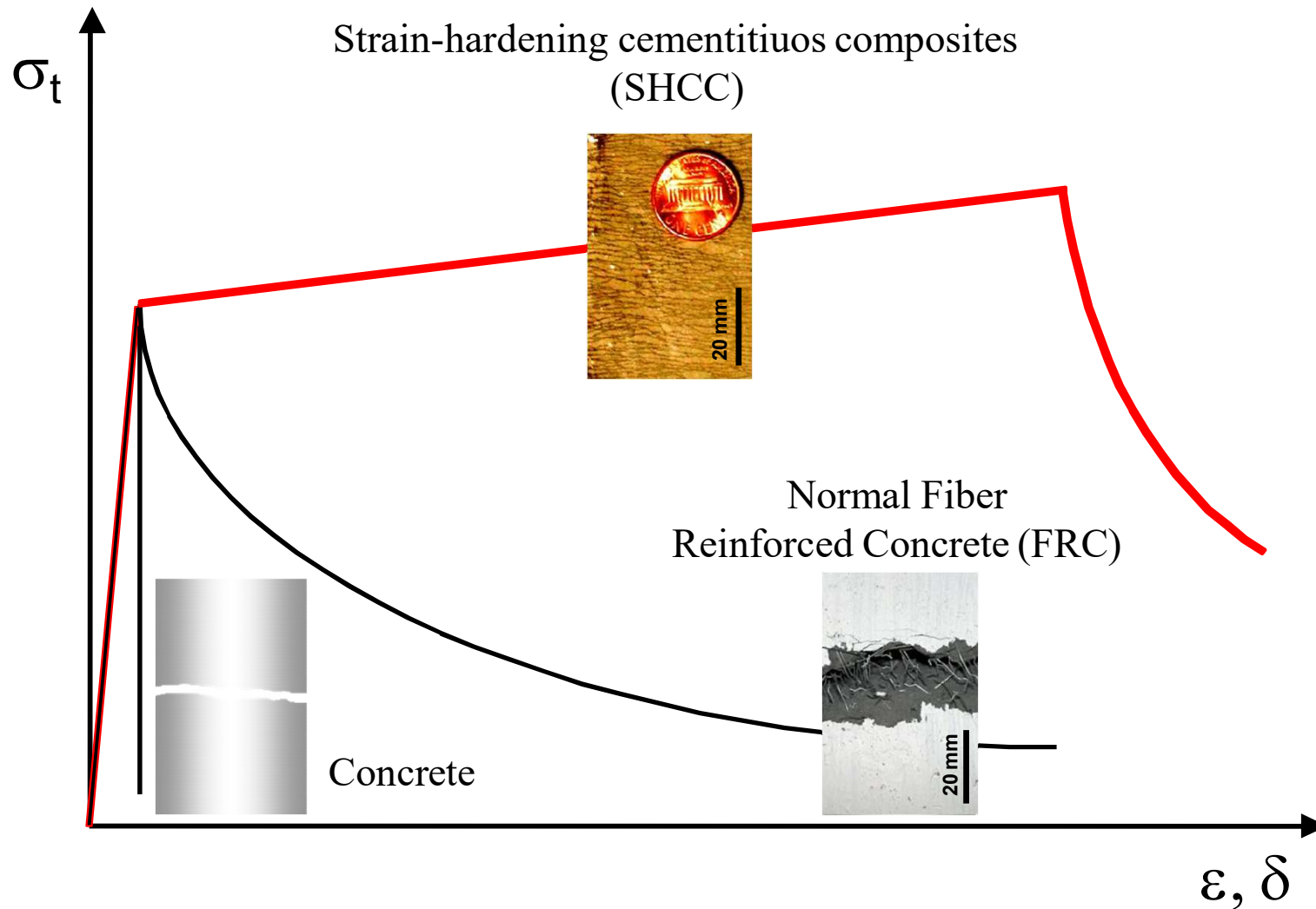
Range of Proportions for Normal Weight FRC



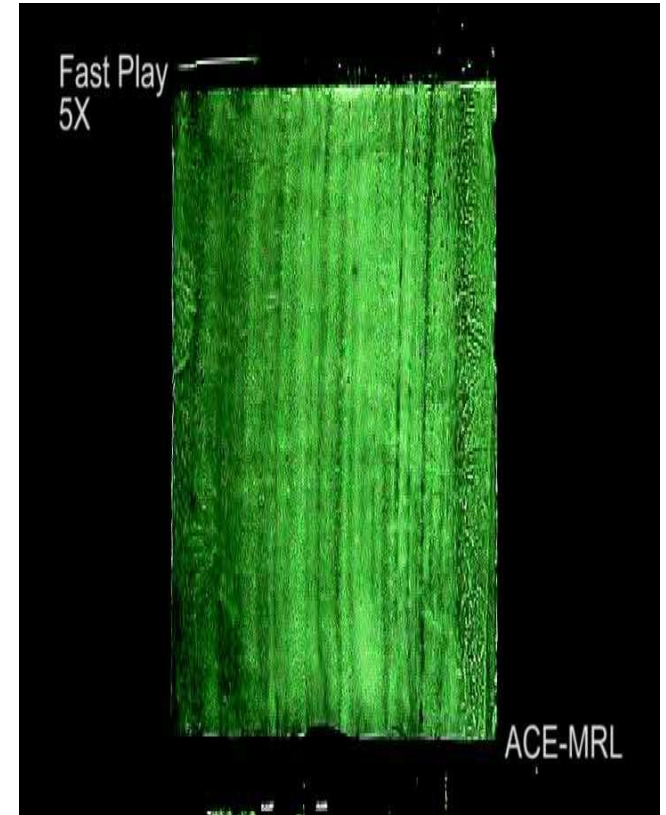
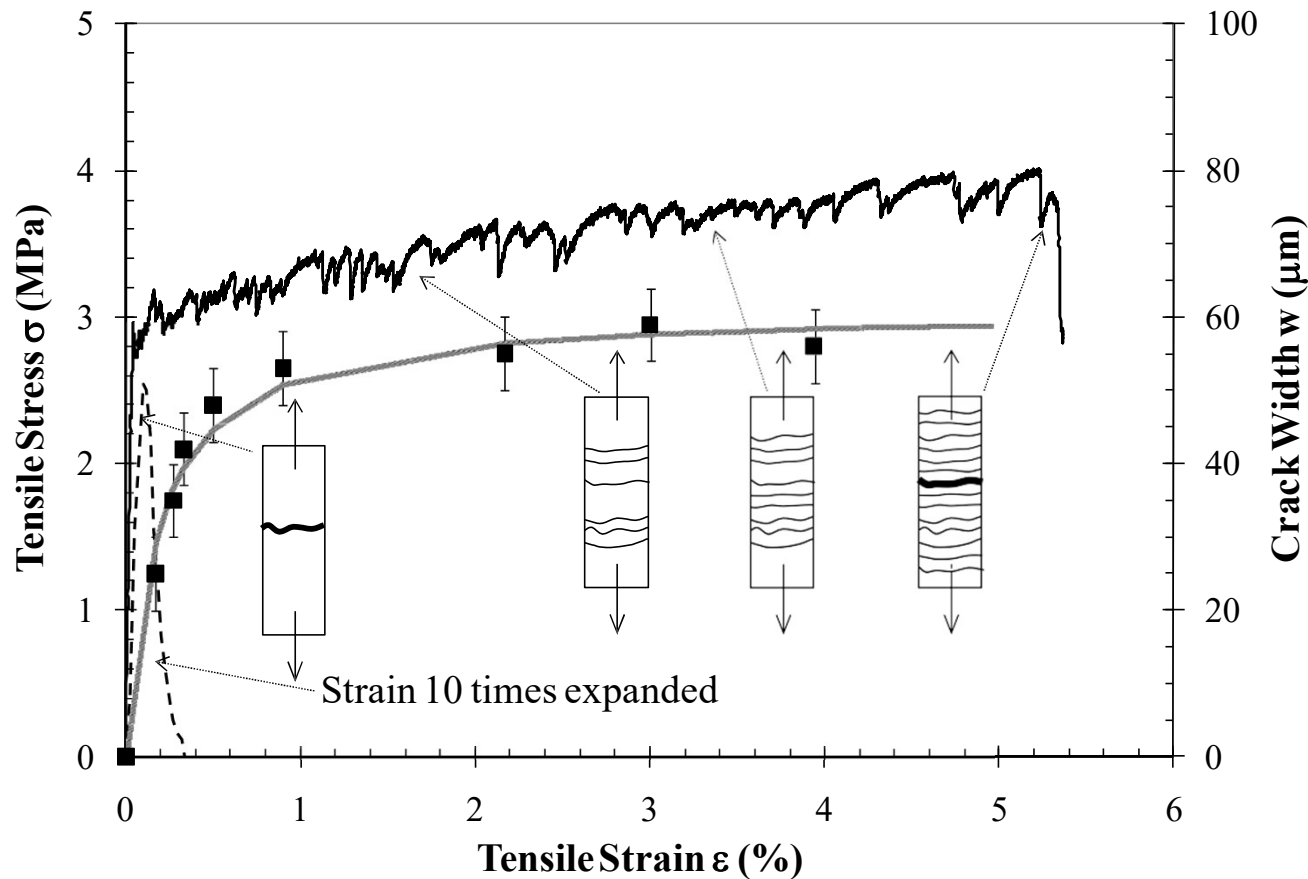
	Mortar	3/8 in. Maximum aggregate	3/4 in. Maximum aggregate
Cement (lb/yd ³)	700-1200	600-1000	500-900
Water/cement ratio	0.30-0.45	0.35-0.45	0.40-0.50
Percent of fine to coarse aggregate	100	45-60	45-55
Entrained air content (%)	7-10	4-7	4-6
Fiber content (vol. %)			
Deformed steel fiber	0.5-1.0	0.4-0.9	0.3-0.8
Smooth steel fiber	1.0-2.0	0.9-1.8	0.8-1.6
Glass fiber	2-5	0.3-1.2	

Source: ACI Committee 544, Report 544.1R-82, *Concr. Int.*, Vol. 4, No. 5, p. 16, 1982.

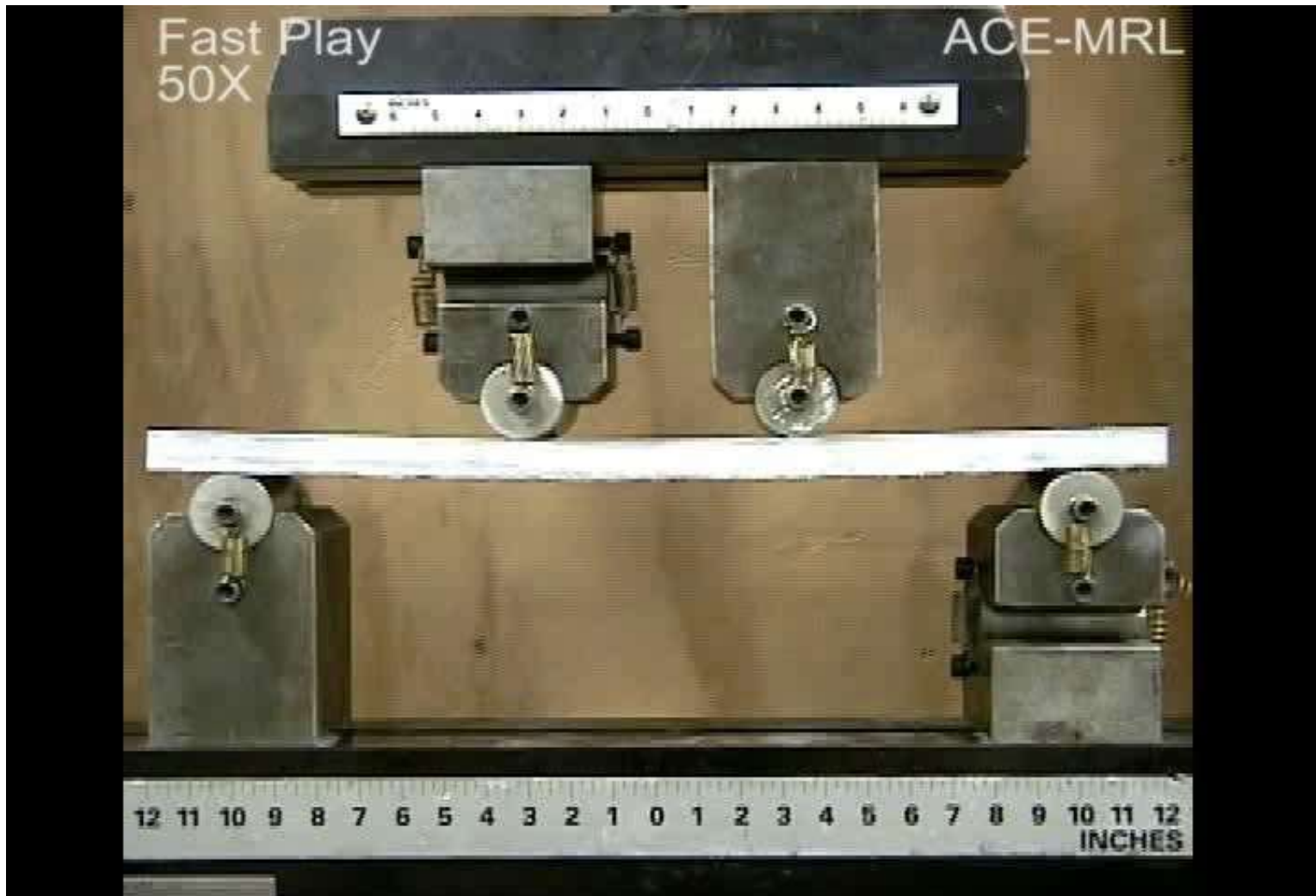
Tensile Behavior of Cementitious Materials



Metal-like Behavior of Engineered Cementitious Composites (ECC)



ECC – Bendable Concrete



<https://www.youtube.com/watch?v=TUxXzjan5k>

<https://www.youtube.com/watch?v=X900IT9dd2g>

Self Compacting Concrete (SCC)

- Self compacting concrete also known as self consolidating concrete is a highly *flowable*, *nonsegregating* concrete that can spread into place, fill the formwork and encapsulate the reinforcement, without any mechanical consolidation.





Advantages of SCC

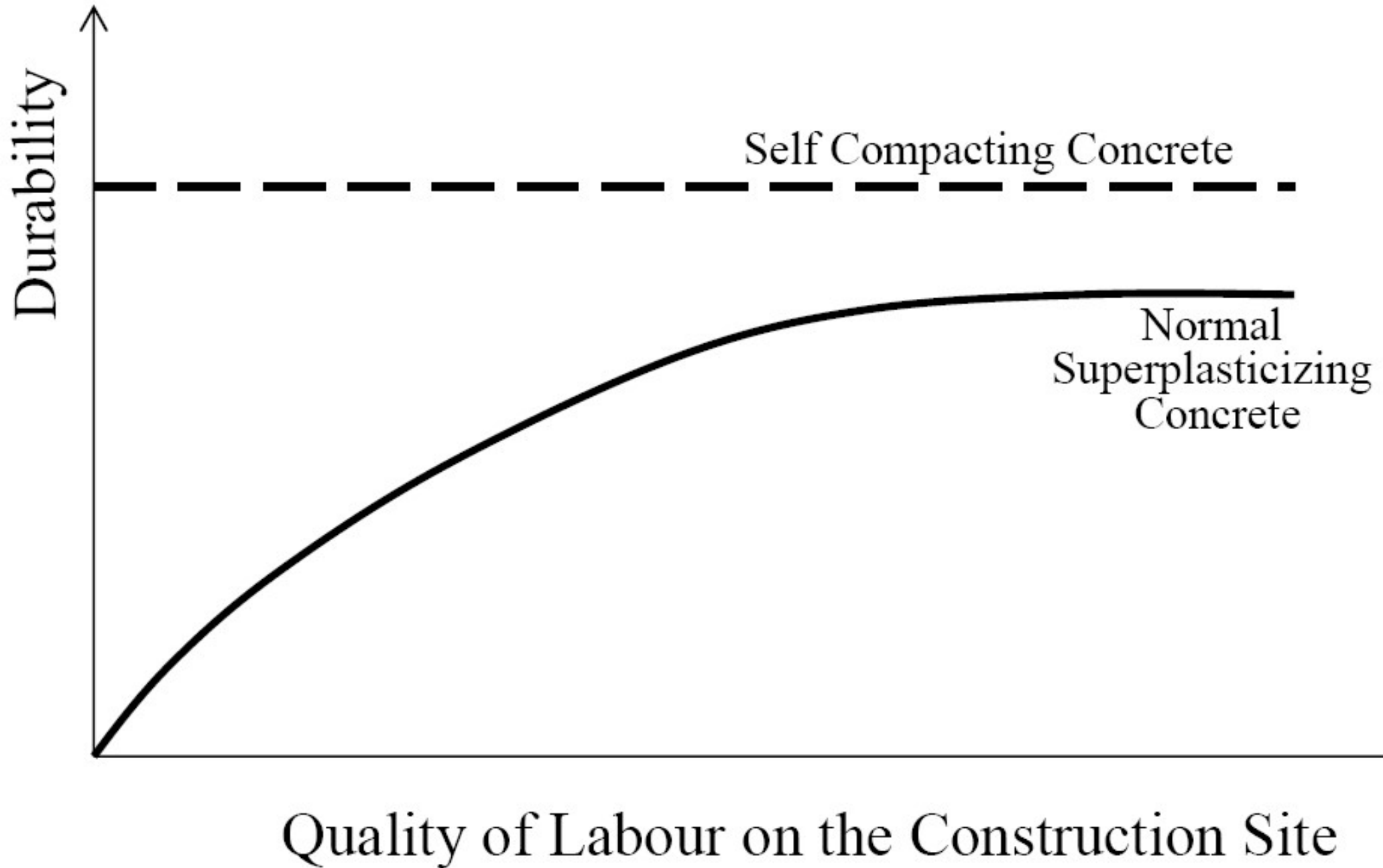
- Increased productivity (faster placement rate, no vibration and less screeding, less manpower)
- Reduced noise due to vibration
- Better surface finish
- Ease of filling restricted sections and hard-to-reach area
- Improved consolidation around reinforcement and bond with reinforcement
- Improved pumpability



Self Compacting Concrete

- Two important properties of SCC are its *flowability* and *stability*, i.e. resistance to segregation
- The most important basic principle for *flowing* and *unsegregable* concretes is the **use of superplasticizer** combined with a relatively high content of powder materials in terms of Portland cement, mineral additions, ground fillers and/or very fine sand.

Quality of Labour vs Durability of Concrete



Economical Advantages of Using SCC

